Representation Collapse in Offline RL: Diagnosis, Eigenvalue Regularisation, and Reward Recovery

Yannick M. Bondo

Abstract. This note examines representation collapse in offline reinforcement learning, focusing on the loss of feature diversity in the shared encoder used by Implicit Q-Learning (IQL) on the D4RL Walker2D-v2 benchmark. We show that the effective rank of the encoder's feature covariance matrix steadily deteriorates during training, compressing Q-value estimates until the policy can no longer distinguish among actions. We argue that this collapse is driven by a feedback loop between bootstrapped value targets and a fixed offline dataset. Eigenvalue regularisation applied to the feature covariance matrix preserves representational diversity and improves both stability and return. We then extend the discussion to reward recovery via adversarial inverse reinforcement learning (AIRL), showing how transferable reward functions can be extracted from expert demonstrations without hand-specified costs. Finally, we outline the structural connection between adversarial reward recovery in RL and preference-based alignment in large language models.

1. Motivation

Offline RL learns policies from a fixed dataset of previously collected transitions, with no further environment interaction. This is the natural setting in domains where online data collection is expensive, risky, or simply impractical, including robotics, autonomous driving, healthcare, and, increasingly, language model alignment.

The central challenge is *distribution shift*. The policy being learned will inevitably query state-action pairs that lie outside the support of the dataset. Standard Q-learning handles this through bootstrapping: the target for $Q(s, a)$ depends on the value of the next state under the current policy. When that next state is out-of-distribution relative to the training data, the bootstrap target becomes unreliable and those errors compound over time.

IQL [1] addresses this by never directly evaluating actions outside the dataset. Instead of computing $\max_{a'} Q(s', a')$, IQL learns a state value function $V(s)$ via asymmetric *expectile regression*:

$$\mathcal{L}_V = \mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[L_\tau^2\Big(Q(s,a) - V(s)\Big)\Big] \tag{1}$$

where the asymmetric loss $L_\tau^2(u) = |\tau - \mathbf{1}_{u<0}|\, u^2$ with $\tau \in (0.5, 1)$ causes V to track the upper tail of the Q-distribution over in-dataset actions. The policy is extracted via *advantage-weighted regression* (AWR):

$$\mathcal{L}_\pi = -\mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[\exp\Big(\frac{Q(s,a) - V(s)}{\beta}\Big)\log\pi(a \mid s)\Big] \tag{2}$$

which upweights high-advantage actions without ever explicitly evaluating actions outside the dataset.

This avoids the most direct form of out-of-distribution extrapolation. It does not, however, resolve a subtler issue: under the statistical constraints of offline training, the *representations* learned by a shared encoder can still degrade. That degradation is the focus of this note.

2. The Observation

All value and policy networks in our IQL implementation share a single encoder $\phi_\theta : \mathcal{S} \to \mathbb{R}^d$. The Q-networks, V-network, and policy all operate on features $z = \phi_\theta(s)$ rather than raw observations.

During training *without* eigenvalue regularisation, we observe a consistent three-stage pattern of degradation:

Stage 1: Feature covariance degenerates. The eigenvalue spectrum of the batch feature covariance $C = \frac{1}{B}\sum_i z_i z_i^\top$ concentrates: a small number of eigenvalues dominate while the rest decay toward zero. The *effective rank* —

$$\mathrm{erank}(C) = \exp\Big(-\sum_i \hat\lambda_i \log\hat\lambda_i\Big), \quad \hat\lambda_i = \frac{\lambda_i}{\sum_j \lambda_j} \tag{3}$$

(the exponential of the Shannon entropy of the normalised eigenvalue distribution) — falls to a small fraction of the embedding dimension d.

Stage 2: Q-value discrimination degrades. As the feature space loses dimensionality, Q-value estimates for different actions at the *same* state converge. The advantage spread $\mathrm{Var}_a[Q(s,a) - V(s)]$ compresses, and the network can no longer distinguish good actions from bad ones.

Stage 3: Policy performance suffers. When advantages collapse, the exponential weighting in AWR (Eq. 2) becomes nearly uniform, and the policy reduces to something close to unweighted behaviour cloning over the entire dataset, including medium-quality trajectories.

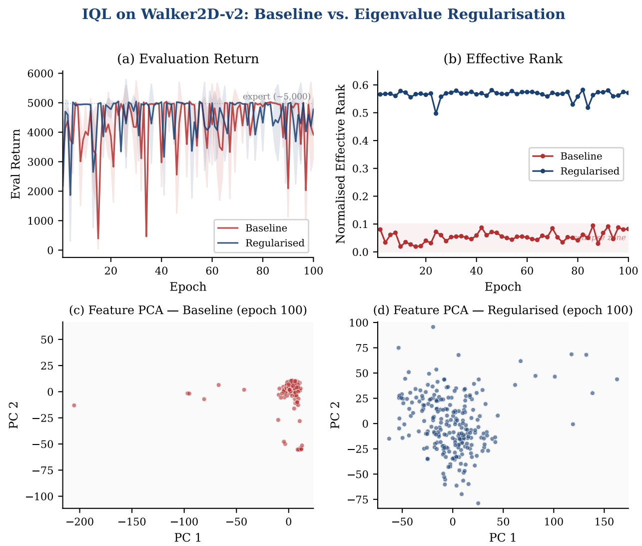


Figure 1: IQL on Walker2D-v2 over 100 epochs. **(a)** Eval return: baseline (red) oscillates between ∼400 and 5,000; regularised (blue) stabilises near expert level (∼5,000). **(b)** Effective rank: baseline collapses to ∼0.05 (13/256 dims); regularised holds at ∼0.57 (145 dims). **(c, d)** PCA of encoder features: baseline collapses into a tight cluster; regularised distributes states across the space.

3. Diagnosis

The collapse has both a *structural cause* tied to IQL's architecture and a *dynamic amplifier* shared by offline value-based methods more broadly.

3.1 Structural cause: Q-to-V compression

IQL's expectile loss (Eq. 1) compresses the action-conditional value $Q(s, a)$ into a state-only summary $V(s)$ through a shared encoder. Whereas Q maps $(s, a) \to \mathbb{R}$, V maps $s \to \mathbb{R}$. In effect, expectile regression asks the encoder to produce features from which V can summarise the *entire* action-conditional value landscape at each state with a single scalar.

The encoder cannot simply preserve features that matter only to Q. If V can explain its targets with fewer dimensions, the shared encoder is pushed to discard the rest. The very features that help Q separate good actions from bad ones at a fixed state are precisely the features that V does not require, because V has already marginalised over actions.

This is the structural predisposition: **IQL's architecture contains an information bottleneck that discards action-conditional structure by design.**

3.2 Dynamic amplifier: the offline feedback loop

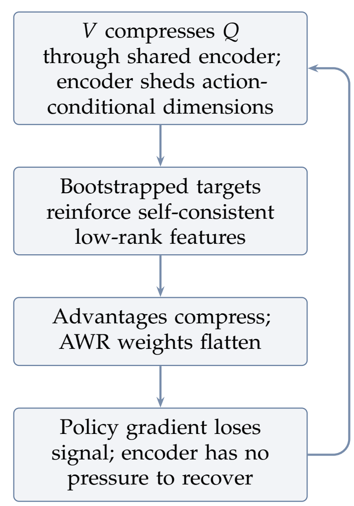
Three factors, all common in offline value-based learning, turn this structural compression into full collapse:

Fixed data distribution. The encoder sees the same finite set of states every epoch. Unlike in online RL, where an improving policy naturally induces a changing curriculum of states, offline training reuses the same transitions indefinitely. There is no exploratory pressure to preserve representational diversity.

Bootstrapped targets. The Q-learning objective $\mathcal{L}_Q = \mathbb{E}[(Q(s,a) - r - \gamma V(s'))^2]$ propagates gradients through the encoder. The target $r + \gamma V(s')$ is computed from features of the same encoder (or a slowly-updating copy), driving the encoder toward representations that make the *current* value estimates self-consistent, rather than representations that preserve fine-grained state-action distinctions.

AWR weight flattening. As advantages compress, the exponential weights $\exp\big((Q - V)/\beta\big)$ in the policy loss flatten toward uniform. The policy gradient then stops providing meaningful directional signal to the encoder, removing one of the last forces that could counteract collapse.

The net effect is a positive feedback loop:



3.3 Implication: scaling without regularisation is wasteful

Representation collapse undermines the basic premise of scaling. A wider network is supposed to learn richer features and support better downstream performance. But if training dynamics compress the feature covariance into only a handful of active dimensions, extra width simply adds unused capacity. A 256-dimensional encoder with effective rank ∼0.03 is effectively using only ∼8 dimensions; doubling the width to 512 would not solve the problem if the same compression mechanism remains in place. The model would pay for additional capacity without gaining additional feature diversity.

Eigenvalue regularisation restores that scaling contract. By preventing the covariance from degenerating, it allows additional encoder capacity to translate into additional feature diversity rather than additional dead dimensions.

This failure mode is not unique to IQL — the same mechanism has been observed across offline RL methods [2] — but IQL's Q-to-V compression makes it structurally predisposed.

4. The Fix: Eigenvalue Regularisation

We add a regularisation term that penalises low-rank feature covariance matrices. Given a batch of encoder outputs $Z \in \mathbb{R}^{B\times d}$ (zero-centred, unit-variance-normalised per dimension), we form the regularised covariance $C_\varepsilon = \frac{1}{B}Z^\top Z + \varepsilon I$ and minimise the negative log-determinant:

$$\mathcal{L}_{\text{reg}} = -\alpha \sum_{i=1}^{d} \log(\lambda_i + \varepsilon) \tag{4}$$

where $\lambda_1, \ldots, \lambda_d$ are the eigenvalues of C_ε. This is equivalent to $-\alpha \log\det(C_\varepsilon)$ and admits a simple interpretation: it penalises eigenvalues that approach zero and therefore discourages collapse into a low-rank subspace. We use $\alpha = 5.0$, $\varepsilon = 10^{-3}$.

The loss is added to the Q-learning objective and backpropagated through the shared encoder. It requires one eigendecomposition per batch of a $d \times d$ matrix (d=256), which is negligible at this scale.

4.1 Encoder Architecture

The shared encoder ϕ_θ is a three-stage network:

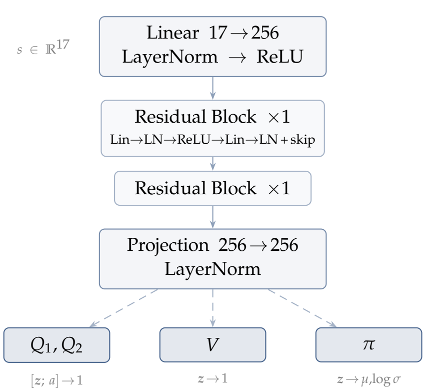


Figure 2: Shared encoder architecture. All downstream heads (Q_1, Q_2, V, π) operate on the same feature vector $z = \phi_\theta(s)$. LayerNorm at each stage stabilises feature scale under competing gradient pressures.

The Q-networks take the concatenation $[z; a]$; the V-network takes z; the policy maps z to the mean and log-standard-deviation of a tanh-squashed Gaussian. Twin Q-networks with minimum aggregation provide conservative estimation. The V-network has a target copy updated via Polyak averaging ($\tau_{\text{target}} = 0.005$).

4.2 Experimental Setup

Dataset. D4RL `walker2d-medium-expert-v2`: 2 M transitions (2,191 episodes) from a mixture of medium and expert policies. Observations are 17-dimensional (joint angles, velocities, torso height and angle); actions are 6-dimensional continuous torques. Expert-level return on this task is ~5,000.

Hyperparameters. Expectile τ=0.7, AWR temperature β=3.0, discount γ=0.99, batch size 256, learning rate 3×10^{-4} for all networks, gradient clip 10.0. Observations normalised to zero mean and unit variance; rewards scaled by 0.1. Seed 42 for both runs.

Controlled comparison. We compare two runs that are identical in every respect except for the regularisation term:

- **Baseline:** IQL with LayerNorm + residual encoder, *no* eigenvalue regularisation.
- **Regularised:** Same, with the regularisation loss (Eq. 4, α=5.0, ε=10^{-3}) added to the Q-learning objective.

Evaluation uses 5 rollouts per epoch with a maximum of 1,000 steps. Representation diagnostics logged every 2 epochs.

4.3 Results

Metric	Baseline	Regularised
Eval return (mean ± std)	4,312 ± 917	4,565 ± 623
Effective rank (mean)	0.055	0.566
Active dimensions (of 256)	~13	~144
Top eigenvalue / trace	0.363	0.019
Off-diagonal correlation	0.357	0.054
Epochs below 3,000	7 / 100	3 / 100
Catastrophic (<1,000)	2 / 100	0 / 100

Table 1: Summary over 100 training epochs. Expert-level return on Walker2D-v2 is ~5,000. Representation metrics are averages over all logged checkpoints. The baseline encoder uses ~5% of its capacity; the regularised encoder uses ~57% — a 10× improvement in representational diversity.

The representation collapse is unambiguous: the regularised run shows a 10× higher effective rank, a 19× lower eigenvalue concentration, and a 7× reduction in feature redundancy (Figure 1b–d). The baseline encoder allocates ~94% of its 256 dimensions to effectively dead features.

The performance consequence is *instability*, not total failure. The baseline can spike to expert-level returns (>5,000) but cannot sustain them — it crashes to <500 twice during training. The regularised run locks in near expert level after epoch 5 and maintains consistent high performance, with zero catastrophic failures.

Critically, the stability gap appears *from the outset*, not only late in training. The regularised encoder maintains ~0.57 effective rank from epoch 2; the baseline is already at ~0.08 by epoch 2 and continues to deteriorate. The regularisation does not merely "fix" a late-stage problem — it prevents the collapse from ever developing.

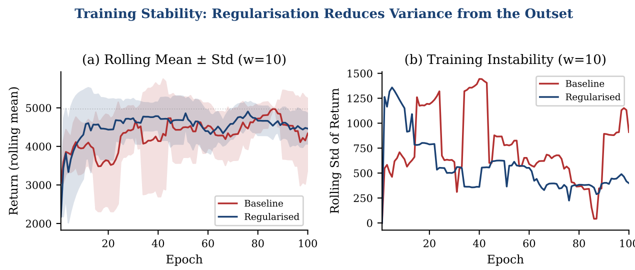


Figure 3: Rolling statistics (window=10 epochs). **(a)** The regularised run's rolling mean stays near expert level with tight variance; the baseline is persistently noisy. **(b)** Rolling standard deviation: the baseline's instability never resolves, while the regularised run stabilises early.

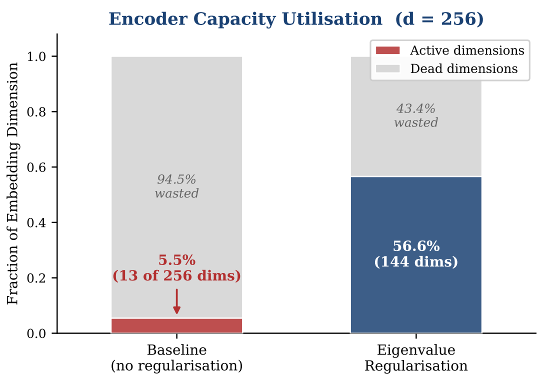


Figure 4: Encoder capacity utilisation. The baseline wastes 94.9% of its embedding dimensions; the regularised encoder uses 56.7%. Without regularisation, scaling the encoder width would produce additional dead dimensions rather than richer representations.

5. Reward Recovery via Adversarial IRL

The previous sections assume access to a known reward function. In many practical settings, however, the reward is unavailable. Behaviour cloning sidesteps that problem by directly imitating expert actions, but it is brittle: errors compound at test time, and the learned policy may fail in situations that were not represented in the demonstrations.

Inverse RL recovers the *reward function* that the expert is implicitly optimising. Once recovered, that reward can be used to train a new policy via standard RL, and that policy can generalise beyond the demonstrations because it has access to the underlying objective.

We use **Adversarial IRL** (AIRL) [3], which frames reward recovery as a discriminator game between expert behaviour and the current policy. The discriminator's learned function decomposes as:

$$f_\theta(s, a, s') = r_\theta(s, a) + \gamma\, V_\theta(s') - V_\theta(s) \tag{5}$$

where r_θ is a learned reward network and V_θ is a shaping potential. The discriminator output is:

$$D_\theta = \frac{\exp(f_\theta)}{\exp(f_\theta) + \pi(a \mid s)} \tag{6}$$

At convergence, r_θ recovers the true reward up to a constant, independent of both the shaping potential and the policy, which makes the learned reward transferable across policies and settings.

Expert dataset curation. From the full D4RL dataset (2,191 episodes), we selected the top 10% by return after filtering for physical plausibility: forward velocity $> 0.5\,\text{m/s}$, torso angle $< 0.3\,\text{rad}$, height $> 1.0\,\text{m}$. This yielded **157 episodes** (157,000 transitions, mean return ~4,971), producing a clean expert set analogous to curating high-quality human demonstrations for reward learning.

Training. The AIRL discriminator was trained adversarially against a SAC policy that collected online rollouts. The discriminator uses the full decomposition (Eq. 5) with separate reward and value heads, stabilised by gradient penalty (λ=50).

Convergence signature. The discriminator follows a characteristic trajectory. Early in training, when neither the discriminator nor the policy has learned much, accuracy sits near chance (~50%). As the discriminator learns to separate expert from policy transitions, accuracy climbs to ~90%. Then, as the policy improves under the learned reward and begins to match the expert's behaviour distribution, the discriminator's task becomes harder and accuracy declines, eventually settling near ~58%. That is the expected equilibrium once the policy and expert distributions are close, though not perfectly matched. This rise-then-fall pattern is the clearest sign that adversarial reward recovery is working as intended: the discriminator first learns the distinction, and the policy then improves enough to erase much of it.

6. Connection to Preference-Based Alignment

The reward recovery problem in AIRL and the alignment problem in large language models can be framed in the same way: given examples of preferred behaviour, infer a reward signal that captures *why* the behaviour is preferred without manually specifying the objective.

In AIRL, the discriminator implicitly represents reward through the separation $f_\theta - \log \pi(a \mid s)$. In Direct Preference Optimisation (DPO) [4], the implicit reward under the Bradley-Terry preference model is:

$$r(x,y) = \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \tag{7}$$

Both formulations absorb reward into the learning objective itself: AIRL through the discriminator, and DPO through the policy loss. In both cases, this avoids the additional instability that can arise when reward learning is separated from policy optimisation. The Bradley-Terry model that connects pairwise preferences to latent reward therefore plays a role analogous to the maximum-entropy IRL objective that links demonstrations to latent reward.

The representation collapse studied in Sections 2–4 is relevant here as well. DPO and related methods (GRPO, IPO) fine-tune a pretrained model under a KL penalty against a reference policy. When that penalty is too weak, the policy drifts far from the reference, and the implicit reward becomes dominated by the policy's own likelihood rather than genuine preference signal — a mode of degradation analogous to the self-consistent but uninformative representations that characterise collapse in offline RL.

We verified this empirically by fine-tuning Qwen2.5-0.5B-Instruct on the Anthropic HH-RLHF preference dataset using DPO at $\beta \in \{0.1, 0.5\}$ (Figure 5). The weak-penalty run (β=0.1) exhibits greater policy drift from the reference model throughout training, while the stronger penalty (β=0.5) keeps the policy closer to the reference. The tradeoff is the same one that appears in representation collapse: if optimisation is left too weakly constrained, the system can settle into a solution that is internally self-consistent but poorly aligned with the signal we actually care about.

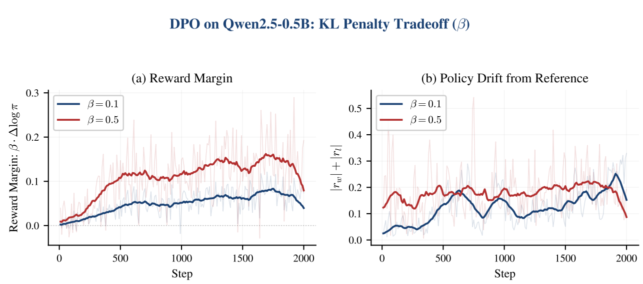


Figure 5: DPO fine-tuning of Qwen2.5-0.5B-Instruct on Anthropic HH-RLHF. **(a)** β=0.5 produces larger reward margins because β scales the log-ratio. **(b)** β=0.1 allows more policy drift from the reference — the implicit reward is increasingly dominated by the policy's own likelihood rather than genuine preference signal.

Summary

1. **Observation.** In offline RL, feature representations can collapse during training: effective rank falls, Q-values lose discrimination, and policy quality becomes unstable.
2. **Diagnosis.** The collapse is driven by a positive feedback loop involving bootstrapped targets, a fixed dataset, and shared encoder gradients.
3. **Fix.** Eigenvalue regularisation ($\mathcal{L} = -\alpha \sum \log \lambda_i$) on the feature covariance preserves representational diversity with negligible overhead.
4. **Reward recovery.** AIRL extracts transferable reward from expert demonstrations via $f(s,a,s') = r(s,a) + \gamma V(s') - V(s)$.
5. **Alignment connection.** Adversarial reward recovery and preference-based alignment (DPO) share the same basic structure: both infer latent reward from observed behaviour without explicit reward engineering.

References

[1] I. Kostrikov, A. Nair, S. Levine. *Offline Reinforcement Learning with Implicit Q-Learning*. ICLR, 2022.

[2] A. Kumar, A. Agarwal, T. Ma, A. Courville. *DR3: Value-Based Deep Reinforcement Learning Requires Explicit Regularization*. ICLR, 2022.

[3] J. Fu, K. Luo, S. Levine. *Learning Robust Rewards with Adversarial Inverse Reinforcement Learning*. ICLR, 2018.

[4] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. Manning, C. Finn. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. NeurIPS, 2023.